

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<answer>A. Radial artery</answer>

<think>Reasoning process here :

- 1 . The anatomical snuff box is a region located on the radial , dorsal aspect of the hand , bounded by the tendons of the abductor pollicis longus , extensor pollicis longus , and extensor carpi radialis brevis .
- 2 . The radial artery is located within this region , running through the anatomical snuff box and traveling laterally to the forearm .
- 3 . Given that the teenager has a penetrating stab wound in the anatomical snuff box , the radial artery is the most likely to be damaged , leading to the excessive bleeding .
- 4 . The ulnar artery is located on the medial aspect of the forearm and is not in the anatomical snuff box , making it an unlikely cause of the bleeding .
- 5 . The Princeps pollicis artery , also known as the proper palmar digital artery , is a branch of the radial artery , but it's not the main artery responsible for the bleeding in the anatomical snuff box .
- 6 . The brachial artery is the main artery of the upper limb , but it's located deeper in the arm and not in the anatomical snuff box , making it an unlikely cause of the bleeding .
- 7 . Therefore , the radial artery is the most likely to be damaged due to the location of the wound and the anatomy of the area .
- 8 . Blood flow to the hand from the radial artery is crucial , and damage to this artery can lead to excessive bleeding in the hand .</think>